Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			(i) (graphite) anode (1) (ii) Molten aluminium (1) (iii) (graphite) cathode (1)	3			3		
	(b)			**2** $Al_2O_3 \rightarrow$ **4 Al** (1) + **3 O_2** (1) one mark for balancing (1) formulae must be correct to obtain balancing mark		3		3		
	(c)			Any 3 × (1) from: At cathode/negative electrode – aluminium <u>ions</u> gain (three) electron(s) (1) (Is reduced) and forms aluminium <u>atoms</u> (1) At anode / positive electrode – oxide (ions) lose electrons / oxidized (1) Forms oxygen (molecules) (1) Allow for (1) Aluminium ions gain electrons and oxide (ions) lose electrons	3			3		
	(d)	(i)		Any 2 × (1) from: Less cost of transport of Al ore to the UK (1) Lower cost of labour (1) no {mining/extraction} aluminium (ore) is required (1) less electrical energy needed to recycle / lower energy cost (1) do not accept 'less cost' unless qualified	2			2		